



BALL STATE
UNIVERSITY

Module 11 Lecture - The Chi-Square Distribution

Introduction to Statistical Methods

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1 Overview and Introduction

1.1 Textbook Learning Objectives

- Interpret the chi-square probability distribution as the sample size changes.
- Conduct and interpret chi-square goodness-of-fit hypothesis tests.
- Conduct and interpret chi-square test of independence hypothesis tests.
- Conduct and interpret chi-square homogeneity hypothesis tests.

1.2 Instructor Learning Objectives

- Appreciate how the chi-square family of tests largely depart from the previous mean-based testing in t-tests
- Understand what scenarios each form of the chi-square test may be appropriate to
- See how the chi-square family of tests add to our toolbox of inferential tests

1.3 Introduction

- The χ^2 distribution, also known as **chi-square distribution**, or chi-squared distribution is a unique tool that can be _____ in inferential testing
 - Much like the _____, the chi-square distribution serves a purpose of _____
- There are 3 primary _____ for the chi-square distribution:
 - Goodness-of-fit: to compare if data is distributed as _____
 - Test of independence: to compare if two _____ are independent
 - Test of homogeneity: to test if two distribution are the _____ as one another

! Important

Though not explicitly mentioned in the book, the chi-square distribution is particularly well-suited to helping us make sense of categorical data - but that is not the ONLY way it can be used

2 Basics of the Chi-Square Distribution

2.1 Formula & Notation

 Discuss: Review: Write out the notation for Student's t-distribution

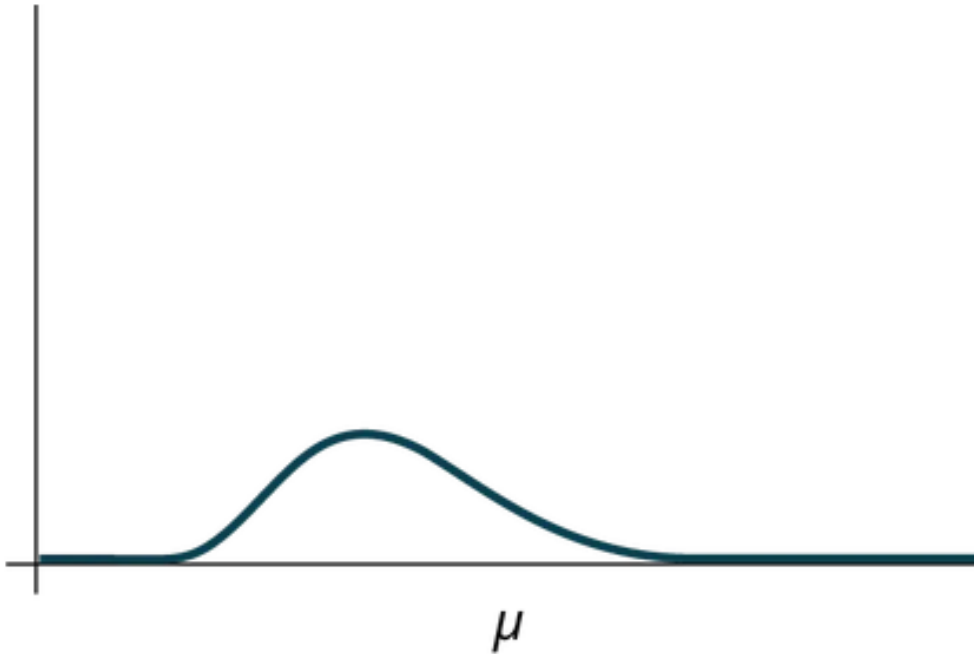
- Formula for χ^2 distribution:
 - $X \sim \chi_{df}^2$
 - Where:
 - * df : degrees of _____
 - * X : a _____ variable (much like other distributions, this can really be any _____ letter, X is just convention)
 - * χ^2 : arbitrary notation for the chi-square distribution

2.2 Distribution Characteristics

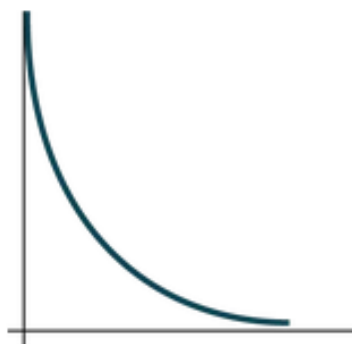
 Discuss: Review: Think back to our initial discussions on the uniform and exponential distributions - write out the formulas for the find long-term mean and standard deviation in those distributions?

- Characteristics of the distribution:
 - $\mu = df$
 - $\sigma = \sqrt{2(df)}$
 - The curve is _____ and skewed to the right
 - The chi-square _____ for each df

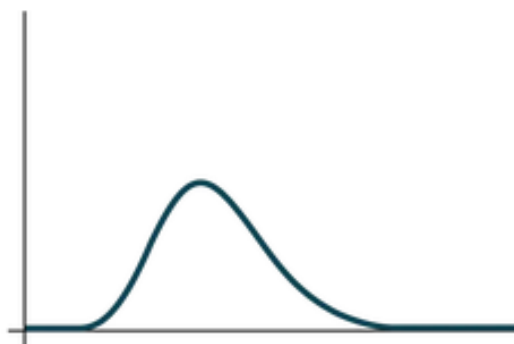
- The test statistic for any test is always _____ than or equal to 0
- When $df > 90$, the chi-square distribution is approximately normal
- The μ is located to the near right of the “peak” of the curve



- The χ^2 distribution thus _____ dependent on the degrees of freedom

 $df = 2$

(a)

 $df = 24$

(b)

2.3 The df for Chi-square Distribution

- So how does one _____ the important df for this distribution?

“I don’t mind not knowing. It doesn’t scare me.” — Richard P. Feynman

- Depends on which of the 3 use cases we are _____

 Discuss: Try explaining, in your own words, what degrees of freedom are again?

 Important

The fact that the chi-square distribution changes dependent on the df is a unique change compared to the normal or t-distributions!

3 Goodness-of-Fit Test

3.1 Introduction

- The **goodness-of-fit test** is an application where the _____ distribution is used to determine whether data “fits” a certain distribution

 Discuss: Why might we want to know if the data we have fit a certain distribution? hint: think about 'assumptions'

3.2 Formula

$$\chi^2 = \sum_k \frac{(O - E)^2}{E}$$

$$df = k - 1$$

- Where:

- O = **observed values** are those that we actually _____ from our sample
- E = **expected values** are those that we _____ our observed values to be if they follow the same distribution, i.e., if the _____ hypothesis is true
- k = the number of different _____, you can think of this as the “pairs” of observed and expected data points

 Discuss: A chi-square statistic characteristic is always equal to or greater than 0; explain why this characteristic is true by looking at the formula

 Important

Due to some mathematical oddities, don't use this unless expected values are at least 5 or greater in each category - can inflate type I error

3.3 Null and Alternative Hypotheses Testing

 Important

Null and alternative hypotheses can and should be stated when doing any and all inferential testing! Still applies here

- Hypothesis pair:
 - Null hypothesis: The _____ values/proportions are the same as the _____ values

- Alternative hypothesis: The _____ values/proportions are different than the _____ values
- Goodness-fit-tests are usually done in _____ manner, as a greater statistic suggest a greater divergence from expected distribution

 Discuss: Draw a normal curve and shade it in such a manner to show a right-tailed test

3.4 Application Example

- I have a class of 20 students, who I expect will have a uniform distribution of As, Bs, Cs, Ds, and Fs
 - $E : \{4, 4, 4, 4, 4\}$
 - $O : \{2, 4, 12, 1, 1\}$
- Hypothesis pair:
 - H_0 : the observed letter grades fits the expected distribution of 4 of each A, B, C, D, F
 - H_A : the observed letter grades do not fit the expected distribution of 4 of each A, B, C, D, F

Grade	Expected (E)	Observed (O)	(O – E)	χ^2 / E
A	4	2	-2	$(-2)^2 / 4 = 1.00$
B	4	4	0	0.00
C	4	12	8	$8^2 / 4 = 16.00$
D	4	1	-3	$(-3)^2 / 4 = 2.25$
F	4	1	-3	$(-3)^2 / 4 = 2.25$
Total	20	20	—	21.50

- $\chi^2 = 21.50, df = 5 - 1 = 4$
 - p to be determined by computer and compared against arbitrary α

 Discuss: Do the same process as demonstrated above, but now using 1,1,1,2,15 as the observed data

4 Test of Independence

4.1 Introduction

- The **test of independence test** is used when we want to determine if two factors or _____ are independent or not
 - This test will use a **contingency table** that shows _____ of two variable in data

 Discuss: Think back to our probability units, what did 'independent' mean in that context and how did it relate to sampling?

Important

Much like goodness-of-fit, don't use this unless expected values are at least 5 or greater in each category - can inflate type I error

4.2 Formula

$$\chi^2 = \sum_{i*j} \frac{(O - E)^2}{E}$$

$$df = (i - 1) * (j - 1)$$

- Where:
 - O : observed values
 - E : expected values*
 - i : number of _____ in the data
 - j : number of _____ in the data
- Expected value (E) for each cell of the table is calculated as:
 - $E = \frac{\sum_{row} * \sum_{column}}{\sum_{total}}$

4.3 Null and Alternative Hypotheses Testing

- Hypothesis pair:
 - Null hypothesis: The two factors are _____
 - Alternative hypothesis: The two factors are _____

4.4 Application Example

- I have a class of 40 students, half of whom are female and half are male, and below are their counts of As, Bs, Cs, Ds, and Fs
 - $F : \{6, 5, 3, 5, 1\}$
 - $M : \{3, 4, 4, 7, 2\}$
- Hypothesis pair:
 - H_0 : The two factors of sex and grade are independent
 - H_A : The two factors of sex and grade are dependent
- Contingency Table:

Grade	Female (F)	Male (M)	Total
A	6	3	9
B	5	4	9
C	3	4	7
D	5	7	12
F	1	2	3
Total	20	20	40

- Expected Counts:

Grade	Female (E)	Male (E)
A	$9 \times 20/40 = 4.5$	4.5
B	$9 \times 20/40 = 4.5$	4.5
C	$7 \times 20/40 = 3.5$	3.5
D	$12 \times 20/40 = 6.0$	6.0
F	$3 \times 20/40 = 1.5$	1.5

- Chi-square statistic:

Grade	Female $(O-E)^2/E$	Male $(O-E)^2/E$
A	$(6-4.5)^2/4.5 = 0.50$	$(3-4.5)^2/4.5 = 0.50$
B	$(5-4.5)^2/4.5 = 0.06$	$(4-4.5)^2/4.5 = 0.06$
C	$(3-3.5)^2/3.5 = 0.07$	$(4-3.5)^2/3.5 = 0.07$
D	$(5-6.0)^2/6.0 = 0.17$	$(7-6.0)^2/6.0 = 0.17$
F	$(1-1.5)^2/1.5 = 0.17$	$(2-1.5)^2/1.5 = 0.17$
Total $\chi^2 = 1.94$		

- $df = (5 - 1)(2 - 1) = 4$, p to be calculated by computer

 Discuss: Do the same process as demonstrated above, but now using F: 1,1,1,2,15 and M: 2,2,2,10,4 as the observed data

5 Test of Homogeneity

5.1 Introduction

- Whereas the _____ test can tell us whether data matches a specified, _____ distribution, we have a different test for testing whether two different sets of data follow the same, unknown distribution
 - We call these are **test for homogeneity**

 Discuss: In what hypothesis testing scenarios might we want to know whether two sets of data seem to have equal distributions? hint: think independent-samples t-test

5.2 Formula

Important

The chi-square statistic here is calculated the same as with the test of independence, but the df is different!

$$\chi^2 = \sum_{i*j} \frac{(O - E)^2}{E}$$

$$df = (i - 1) * (j - 1)$$

- Where:
 - O : observed values
 - E : expected values*
 - i : number of _____ in the data
 - j : number of _____ in the data
 - The _____ in this setup should be the categorical variable whose _____ we are interested in
 - The _____ should be the _____ categorical variable, or the two groups we compare against one another

5.3 Null and Alternative Hypotheses Testing

- Hypothesis pair:
 - Null hypothesis: The _____ of the two populations are the same

- Alternative hypothesis: The distributions of the two populations are the same

5.4 Application Example

- I have a group of students, half of whom are female and half are male, and below are their counts of what class they are in:
 - Freshman, Sophomore, Junior, Senior
 - $F : \{6, 14, 8, 12\}$
 - $M : \{5, 7, 7, 8\}$
- Hypothesis pair:
 - H_0 : The distributions of class between male and female students are the same
 - H_A : The distributions of class between male and female students are not the same
- Data setup

Class	Female (F)	Male (M)	Total
Freshman	6	5	11
Sophomore	14	7	21
Junior	8	7	15
Senior	12	8	20
Total	40	27	67

- Expected counts

Class	Expected F	Expected M
Freshman	6.57	4.43
Sophomore	12.54	8.46
Junior	8.96	6.04
Senior	11.94	8.06

- Chi-square calculation

Class	Group	(O)	(E)	$((O-E)^2/E)$
Freshman	F	6	6.57	0.049
Freshman	M	5	4.43	0.074
Sophomore	F	14	12.54	0.170
Sophomore	M	7	8.46	0.252
Junior	F	8	8.96	0.103
Junior	M	7	6.04	0.151

Class	Group	(O)	(E)	$((O-E)^2/E)$
Senior	F	12	11.94	0.0003
Senior	M	8	8.06	0.0004

6 Conclusion

6.1 Recap

- The chi-squared distribution is well suited to dealing with categorical counts and proportions, but the 3 forms of the chi-square distribution have somewhat different application and formulas
- The shape of the chi-square distribution changes based upon the dfs, and the hypothesis testing done with it is usually right-tailed, among other consistent characteristics
- Like with the other distributions we have discussed, be mindful of how the null and alternative hypotheses change

Key Terms

chi-square distribution A unique distribution in which the mean is equal to the degrees of freedom (df) and the standard deviation $\sqrt{2(df)}$ **2**

contingency table A table with a variable on either axis to show counts of occurrence for each level of those variables **8**

expected values Those values which we arbitrarily expect or test against **6**

goodness-of-fit test An inferential statistical test using the χ^2 distribution, tests whether data appears to fit or be the same as a pre-defined distribution **5**

observed values Those which we gather/measure as part of an study or sample **6**

test for homogeneity An inferential statistical test using the χ^2 distribution, tests whether data appears to have the same distribution as another set of data **10**

test of independence test An inferential statistical test using the χ^2 distribution, tests whether data appears to independent of another set of data **8**